

Scalability & Future Plan

Date	02 July 2026
Team ID	
Project Name	Darshan Ease – Temple Darshan Ticket Booking System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how Darshan Ease can be scaled to handle larger user bases, increased data loads, or extended features in the future, and captures the team's roadmap for evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The JWT secret is currently hardcoded directly in authMiddleware.js instead of being read from an environment variable.	Security risk – if the source code is exposed, the secret used to sign/verify tokens is exposed with it.	High
2	The application runs as a single Node.js server instance with a single MongoDB connection, with no load balancing.	Limits how many concurrent users the system can reliably serve as traffic grows.	Medium
3	No automated test suite and no caching layer for frequently accessed data like the temple/product list.	Harder to catch regressions early, and repeated requests hit the database directly, adding avoidable load.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Runs on a single local Node.js/Express server, suitable for a small number of concurrent users.	Deploy to a cloud host (e.g. Render, AWS, or Azure) with the ability to scale horizontally / add load balancing as traffic grows.
2	Data Storage	Single local or free-tier MongoDB instance storing Users, Products, and Orders.	Move to MongoDB Atlas with replica sets, and add sharding if data volume grows significantly.
3	Performance	No caching layer; every request for temple/product data hits MongoDB directly, with no pagination on lists.	Add caching (e.g. Redis) for frequently accessed data and add pagination to the products and orders endpoints.
4	Security	JWT secret is hardcoded in code; no rate limiting on the API.	Move the JWT secret to environment variables, add rate limiting on auth/booking endpoints, and enforce HTTPS in production.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Integrate an online payment gateway (e.g. Razorpay/Stripe) for paid darshan bookings.	1–2 months	Enables real end-to-end paid bookings instead of confirmation-only orders.
Phase 2	Add QR-code based e-tickets and email/SMS booking confirmations.	2–3 months	Faster entry verification at temples and more reliable booking confirmations for users.
Phase 3	Build an admin analytics dashboard for booking trends and slot utilization.	3–4 months	Gives temple administrators visibility into demand to plan slots better.
Phase 4	Add multi-language support and move to a scaled cloud deployment with caching.	4–6 months	Wider accessibility for devotees and better performance under higher load.